

Logan Fouts

Cleveland, Ohio | 440-463-8587 | loganfouts@proton.me | U.S Citizen

Education

Linnaeus University | Vaxjo, Sweden | GPA: 3.34

Sep 2021 – June 2025

Bachelor of Science in Computer Science

- **Concentrations:** Software Technology & Embedded Systems
- **Relevant Coursework:** Data Structures & Algorithms, Software Engineering, AI/ML Fundamentals, Operating Systems, Systems Programming, Web Development

Skills

Hard Skills: PCB Design, Soldering, Microcontrollers, Linux System Administration

Software: Git, Docker, Github actions, KiCad, VS Code, Linux

Programming: Java, Python, C, Assembly, HTML/CSS, Javascript

Activities: Technical Mentor: Created refined study materials as Git repos for students | SFI (Swedish for Immigrants): Student | Study Abroad: Linneuniversitetet, Sweden

Experience

Scale AI | AI Training Programmer | Remote

Oct 2024 - Present

- Optimized AI training workflows implementing corrected solutions in Python and other languages in response to AI outputs.
- Wrote detailed documentation for faults discovered in AI systems and their integration with external services.

Contrib Cal | Founder / Product Developer

Jul 2025 - Present

- Launched an open-source IoT product and software hub that visualizes GitHub activity on a physical LED calendar.
- Secured sponsorship from PCBWay as well as partnership with Elecrow electronics.
- Designed and built the full stack, including a custom PCB designed in KiCAD, Python/Bash firmware, Systemd unit files, and web UI.
- Transformed the project into a commercial product with product launch website and partnership with Elecrow for professional manufacturing.

Freelance Developer | Self-Employed | Chagrin Falls, Ohio

Jul 2024- Present

- Developed full-stack web applications for clients using React, Node.js, and Tailwind.
- Built cross-platform mobile apps with React Native and used requirements engineering for client communication.
- Created custom solutions including a bonsai business website and educational app for kids.

Valley Presbyterian Church | IT Technician / Linux System Admin | Chagrin Falls, Ohio

Jun 2022 – Aug 2022

- Designed and maintained Linux servers for website and document management systems.
- Implemented automated backup solutions with Bash/Cron reducing backup process time by 40%.

Projects

NuPhy Linux: USB HID Communication for Keyboard Control | Open Source Contribution

Sep 2025

Found keyboard I use has no support under linux for controlling the leds as well as keybinds. So, I designed a command line solution.

- Developed a C-based command line utility enabling full control of NuPhy 75 keyboard light and key mappings on Linux through direct USB HID communication.
- Reverse-engineered the keyboard's proprietary USB communication protocol using Wireshark and diff checker to document packet structure and command codes.

MeshBot: Solar Powered Wilderness Safety Device

Current

While hiking Washington's North Cascades, my friends and I realized how valuable instant safety information could be, and how impossible it was to get without cell service or an experienced hiker nearby. I built MeshBot to solve this: a solar-powered AI assistant that lives on the Meshtastic network and answers questions right from the trail.

- Developed an off-grid AI assistant that operates over Meshtastic's LoRa network to provide wilderness safety information without internet or cell service.
- Engineered the full system to be low cost and run on only a low power computer and use a cheap Meshtastic node like a Heltec v3.
- Implemented simple Python based software enabling communication from Meshtastic messages to local AI and back.

Multi-Method NDI Detection Framework | Published Thesis

Mar 2024

While organizing a large family photo collection, I discovered a complex challenge: rather than simple duplicates, most images were near-duplicates with subtle variations, requiring a more sophisticated detection approach.

- Published research on a multi-algorithm detection framework achieving 97% F1-score with significantly reduced level of complexity and computational load.
- Designed layered software architecture combining multiple detection algorithms for enhanced flexibility.